Phase 1 Report | Amazon Reviews’23 / Electronics

Track B: Multimodal Review Analysis | Pilot evidence package | 01 November 2026

Status: Evidence-based pilot report. NOT a final representative analysis or final model benchmark. The source dataset used for this document is the sequential first 1,000 Electronics review records.

1. Problem and research questions

Business/analytical objective: test whether eligible product metadata adds predictive value to review text when estimating the associated 1–5 star rating for previously unseen products. The unit of analysis and prediction is one review, linked to its product using parent_asin.

Q1: How well does review text alone classify the rating? Q2 (selected): Does non-leaking metadata improve a text-only classifier on the same eligible reviews? Q3: Does predictive behavior vary by review length or product review frequency? Q2 was selected because it tests the distinctive multimodal contribution of Track B.

2. Dataset provenance and sampling

Source: Amazon Reviews’23 (McAuley Lab / UCSD), Electronics category review and metadata JSONL archives. Reviews have rating, text, title, images, parent_asin, user_id, timestamp and interactions. Product records can include price, title, category, description and image references. The current study is observational; reviews are user-generated, not controlled experimental data.

Current sampling: first-N sequential selection of 1,000 review rows, followed by scanning metadata for matching parent_asin. Therefore the pilot is not random or representative. The metadata search yielded 275 products matching 297 review rows. Exact archive dates, release version and retrieval timestamps should be recorded locally before submission.

3. Basic audit

Audit metric	Observed pilot
Review rows	1,000
Distinct reviewed products	959
Distinct users	195
Exact duplicate review rows	0
Empty review text	0
Median text length	166 characters
Metadata-matched review rows	297 (29.7%)

The rating is the five-class classification target, not a feature. Distribution: 1 star = 90, 2 = 35, 3 = 65, 4 = 145, 5 = 665. The five-star majority constitutes 66.5%, motivating macro-F1 as the primary metric.

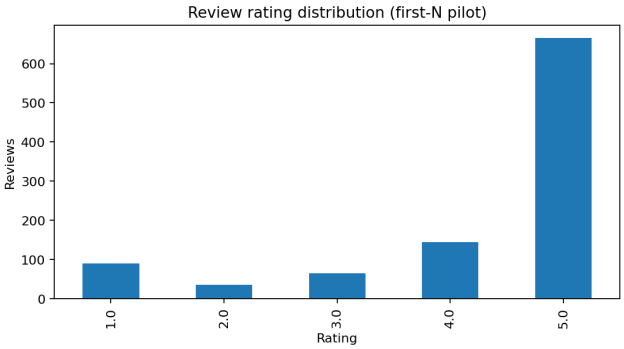


Figure 1. Rating distribution. The 5-star majority dominates this pilot.

4. Exploratory data analysis (EDA)

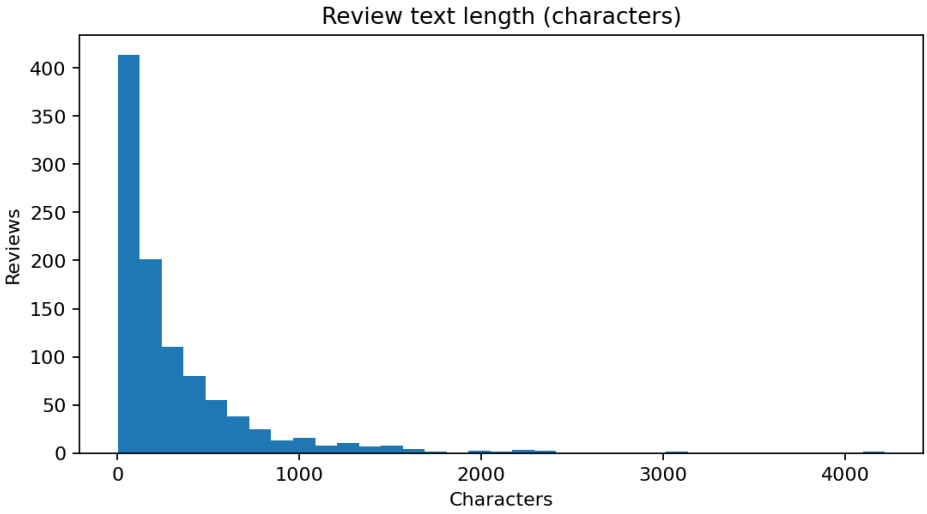


Figure 2. Review character length has a long right tail; median is 166.



Figure 3. Length distributions differ descriptively across rating groups. Outliers may be hidden for readability; no significance test is claimed.

The current pilot contains many products with only one or a few reviews. This limits product-level subgroup inference and makes product-group partitioning especially important. The temporal chart describes only the selected file prefix, not marketplace growth.

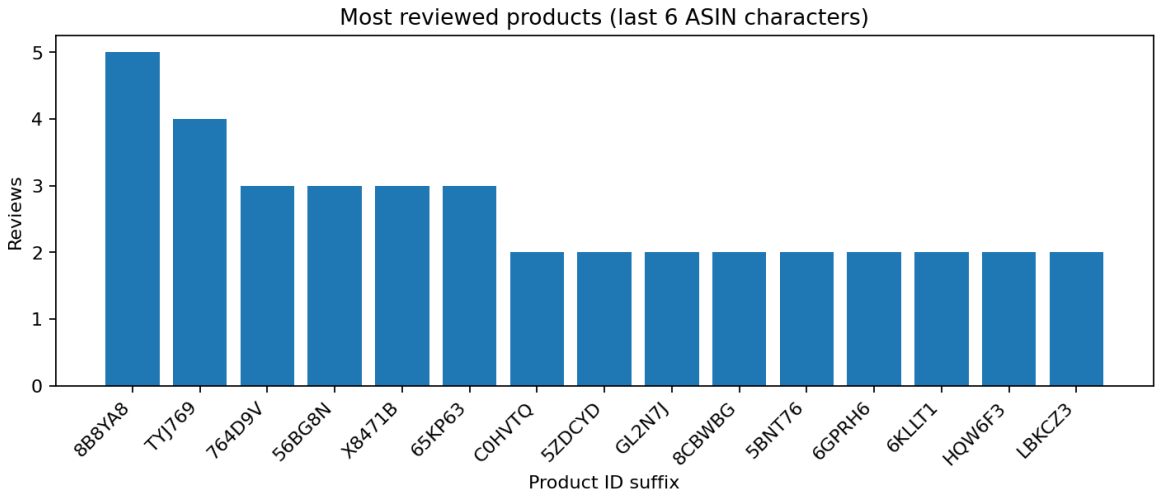


Figure 4. Products with the largest number of reviews in this selected pilot.

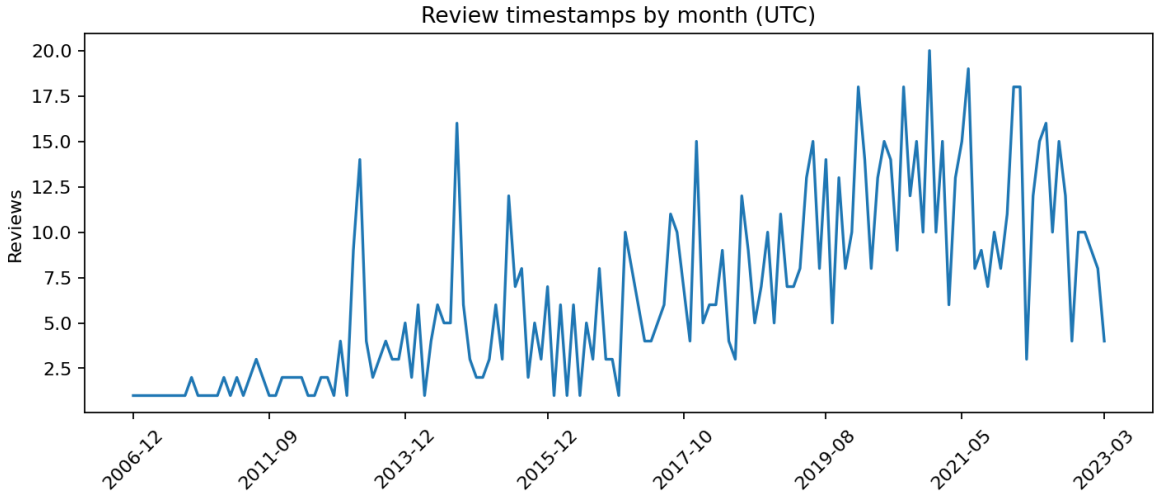


Figure 5. Monthly count of reviews in the sequential pilot; not a population trend estimate.

5. Multimodal coverage and missingness

Actual modality availability among all 1,000 reviews: text 100%, rating 100%, genuine nonempty review-image lists 2%, linked product metadata 29.7%, usable product price 21.2%, product images (references) 29.7%, descriptions 12.7%, categories 29.5%, timestamp 100%, user ID 100%. Product image references do not mean that image bytes have been downloaded.

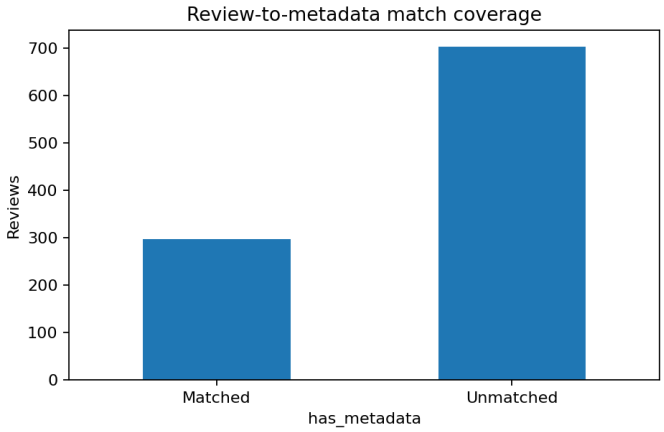


Figure 6. 297 matched and 703 unmatched review rows.

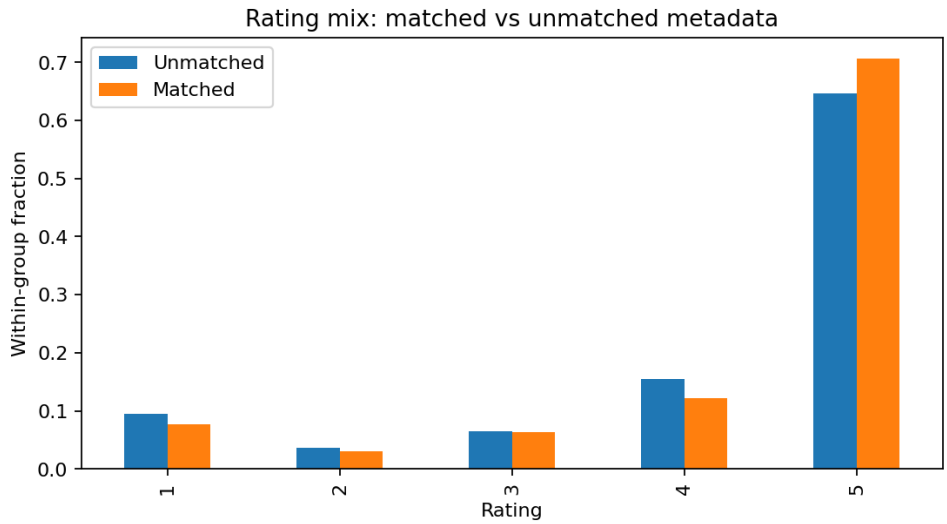


Figure 7. Rating distribution by metadata availability group.

Group	n	Mean rating	5-star %	Median length	Median year
Matched	297	4.350	70.71%	162	2020
Unmatched	703	4.222	64.72%	167	2019

Metadata availability is incomplete and apparently selective. The complete-case subset has a higher observed share of five-star ratings than the unmatched subset. No statistical significance or missing-at-random assumption is established. Direct comparison of models across different subsets would confound the effect of modality availability with the effect of the model.

6. Data quality and leakage risks

Key risks include: (i) using the rating label as a feature; (ii) product average_rating / rating_number aggregations potentially incorporating evaluation labels; (iii) helpful_vote accruing after posting; (iv) repeated or very similar text across splits; (v) user identity overlap; (vi) fitting vectorizers or imputers before splitting; (vii) product metadata reflecting information from later dates. Exclude rating, average_rating, rating_number and helpful_vote from initial feature sets. Learned preprocessing is fitted on training only.

Further required checks: price semantics and invalid entries; extreme lengths and unusual characters; null-vs-empty image records; potential temporal mismatch; metadata availability versus rating/year/text length. Zero exact duplicates is not proof that there are no near-duplicates.

7. Validation protocol

Partition	Review rows	Distinct products	1-star	2-star	3-star	4-star	5-star
Train	702	671	63	21	49	105	464
Validation	150	144	14	5	9	24	98
Test	148	144	13	9	7	16	103

Proposed method: GroupShuffleSplit using parent_asin, random seed 42, approximately 70/15/15. The manifest generator includes a product-disjoint assertion, but independent verification against the saved manifests is pending because the manifests were not supplied for this report. A separate verification script accompanies this document.

Observed user overlap remains nonzero: train/validation = 53; train/test = 49; validation/test = 30. This design estimates transfer to unseen products, not unseen users. A diagnostic random row split had 16 shared product IDs and 69 shared user IDs between train and validation.

Freeze and version the product-disjoint manifests before comparative modeling. No model selection, tuning or repeated evaluation on the test partition. The same matched-row subset and split identity must be used for text-only, metadata-only and combined comparisons; wider-sample text-only results can be reported separately.

8. Baselines: exploratory diagnostics only

Model	Accuracy	Macro-F1	Interpretation
Majority-class Dummy	0.6745	0.1611	High accuracy from dominant class
TF-IDF + Logistic Regression	0.6477	0.3936	Higher class-balanced performance

These outcomes originate from an older exploratory train/validation split (702/298). They are not measured on the current train/validation/test manifests and must not be presented as final benchmarks. No final test performance is reported.

9. Planned analyses and ethics

Later phases: metadata-only, text-only and combined models on identical eligible examples; optional sentence embeddings; TF-IDF/SVD clustering with interpretation and stability; subgroup diagnostics by length and product frequency. Privacy: do not publish raw user IDs or potentially identifying review excerpts. Account for positive-rating dominance, sampling bias and incomplete metadata.

10. Phase 1 submission readiness

Prepared: project card, review/data audits, EDA figures, multimodal availability audit, exploratory baseline, and split summary. Not yet verified: manifest intersections, full provenance/retrieval log, explicit treatment of invalid values, clean-environment reproducibility, and Git tag. Those checks are required before considering Phase 1 formally complete.

Evidence files used: audit.json; eda_summary.json; modality_leakage_summary.json; matched_vs_unmatched.csv; split_summary.json; PROJECT_CARD_phase1_updated.md; seven EDA figures from the project.